

Introduction to Artificial Intelligence

Meaning, Characteristics and Objectives of AI

Meaning of Artificial Intelligence

Artificial Intelligence (AI) is the branch of computer science concerned with building machines and software systems capable of performing tasks that would normally require human intelligence — such as reasoning, learning, perceiving, understanding language, and making decisions.

Key Definitions

- **John McCarthy (1956):** "AI is the science and engineering of making intelligent machines, especially intelligent computer programs."
- **Alan Turing (1950):** A machine can be said to be intelligent if its behaviour, observed through interaction, is indistinguishable from that of a human.
- **Russell & Norvig:** AI can be viewed along two dimensions — thinking vs acting, and humanly vs rationally — giving four categories: systems that think like humans, act like humans, think rationally, and act rationally.

In simple terms, AI enables machines to simulate cognitive functions such as perceiving their environment, learning from experience, reasoning about problems, and taking actions to achieve specific goals.

Characteristics of AI

- **Learning ability:** Improves performance through experience and data (Machine Learning).
 - **Reasoning and problem-solving:** Applies logic and inference to draw conclusions and solve problems.
 - **Perception:** Interprets sensory input such as images, speech, and text (Computer Vision, Speech Recognition).
 - **Natural Language Understanding:** Processes and generates human language.
 - **Adaptability:** Adjusts behaviour in response to a changing environment or new data.
 - **Autonomy:** Performs tasks and makes decisions with minimal human intervention.
 - **Goal-directed behaviour:** Actions are oriented toward achieving defined objectives efficiently.
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Objectives of AI

- To create systems that can perform tasks requiring human-like intelligence, such as recognising patterns and making decisions.
- To automate repetitive, tedious, or dangerous tasks, freeing humans for higher-value work.
- To enhance and augment human decision-making through data-driven insights.
- To enable machines to learn from data and improve performance over time without being explicitly reprogrammed.
- To solve complex, large-scale problems (e.g., in healthcare diagnostics, logistics, climate modelling) beyond unaided human capability.
- To develop natural and intuitive interaction between humans and machines (chatbots, voice assistants).

History and Evolution of AI

Pre-1956: Foundations

- 1943 — Warren McCulloch and Walter Pitts propose a mathematical model of an artificial neuron, laying groundwork for neural networks.
 - 1950 — Alan Turing publishes "Computing Machinery and Intelligence," proposing the Turing Test and asking "Can machines think?"
 - 1951 — First AI-related programs: Christopher Strachey's checkers program and Dietrich Prinz's chess program.
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The Dartmouth Conference (1956)

Widely regarded as the birth of AI as a formal field, the Dartmouth Summer Research Project on Artificial Intelligence was held in 1956 at Dartmouth College, USA.

- **Organisers:** John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon.
 - **Significance:** McCarthy coined the term "Artificial Intelligence" for the proposal. The conference brought together researchers who believed that "every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."
 - **Outcome:** Established AI as a distinct academic discipline and set its long-term research agenda.
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Major Milestones Timeline

Period	Milestone
1956	Dartmouth Conference — AI founded as a field
1958	John McCarthy develops LISP, the dominant AI programming language for decades
1965–66	ELIZA — early natural language chatbot by Joseph Weizenbaum
1969	Shakey the Robot — first mobile robot capable of reasoning about its actions
1970s	Expert systems research begins (e.g., MYCIN for medical diagnosis)
1980s	Commercial boom in expert systems; backpropagation popularised for neural nets
1997	IBM's Deep Blue defeats world chess champion Garry Kasparov
2011	IBM Watson wins the quiz show Jeopardy!
2012	Deep learning breakthrough — AlexNet dramatically improves image recognition
2016	Google DeepMind's AlphaGo defeats Go champion Lee Sedol
2020s	Rise of large language models and Generative AI (GPT, and similar systems)

AI Winters

"AI Winters" refer to periods of reduced funding and interest in AI research, caused by unmet expectations.

First AI Winter (mid-1970s)

- Triggered by the 1973 Lighthill Report in the UK, which criticised AI's failure to achieve its grand promises.
- Machine translation projects failed to deliver on early hype; funding (notably from DARPA) was cut sharply.

Second AI Winter (late 1980s – early 1990s)

- Collapse of the commercial market for expert systems and specialised "Lisp machines."
 - Expert systems proved expensive to maintain and brittle outside narrow domains.
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Resurgence of AI

- Availability of massive datasets ("Big Data") from the internet and digitisation.
- Increased computational power, especially GPUs, enabling large-scale parallel computation.
- Advances in algorithms, particularly deep learning and backpropagation-based neural networks.
- High-profile successes (Deep Blue, Watson, AlphaGo) that restored public and investor confidence.
- Open-source frameworks (TensorFlow, PyTorch) that democratised AI research and development.